AXOLABS METHOD VALIDATION REPORT FOR THE ANALYSIS OF 2439-EXNA IN RAT CSF	
Axolabs Project Number	AN405.0-24
Project Title	Validation of an AEX-HPLC-based method for the analysis of 2439-exNA in rat CSF
Study Director	Nicole Rottmann, Scientist
Test Site	Axolabs GmbH Fritz-Hornschuch-Straße 9 95326 Kulmbach Germany
Sponsor	The Broad Institute, Inc. 75 Ames St Cambridge, MA 02142 USA
Sponsor's Representative	Alissa Coffey The Broad Institute, Inc. 75 Ames St Cambridge, MA 02142 USA
Start Date of Bioanalytical Test Phase	18-Sep-2024
End Date of Bioanalytical Test Phase	25-Sep-2024

SIGNATURE PAGE

Name and Function	Date and Signature
Nicole Rottmann **STUDY DIRECTOR**	17-Dec-2024 [redacted]
Name and Function	Date and Signature
Alissa Coffey **SPONSOR'S REPRESENTATIVE**	18-Dec-2024 [redacted]

COMPLIANCE STATEMENT

This *Validation of an AEX-HPLC-based method for the analysis of 2439-exNA in rat CSF* was conducted in accordance with the procedures described herein.

The method validation was conducted in compliance with *ICH guideline M10 on bioanalytical method validation and study sample analysis* (EMA/CHMP/ICH/172948/2019) and following the rules in the *OECD Principles of Good Laboratory Practice* (Revision from 1997). Due to putative adhesion effects, the stability samples were prepared slightly deviating from the standard procedure described in the ICH M10, as described in Section 9.10.

There were no deviations from the Method Validation Plan and respective SOPs. All work was done according to the Method Validation Plan and respective SOPs.

The Method Validation Report represents an accurate and complete record of the results obtained during the method validation.

Study Director:	Nicole Rottmann	17-Dec-2024 [redacted]
	Name	Date and Signature

QUALITY ASSURANCE STATEMENT

This Method Validation Report titled *Validation of an AEX-HPLC-based method for the analysis of 2439-exNA in rat CSF* has been audited.

The reported results accurately reflect the raw data generated during this method validation.

As the bioanalytical testing is of short duration, routine and repetitive, Axolabs GmbH applies a process-based inspection program. The process-based inspection relevant to the activities performed during this method validation is listed below.

Table 1: Quality Assurance Inspections

Type(s) of Inspection(s)	Inspected Activities	Date(s) of the Inspection(s)	Date(s) of Reporting and Recipient(s)
Process-based (Report No.: II-24-003)	• Incoming Goods Check, storage, and labeling of study samples • Incoming Goods Check, storage, and labeling of reference standard, probe, blank matrix • Labeling and preparation of reagents/ solutions • Labelling and preparation of reference standard stock solution • Preparation of calibration standards and QC samples • Sample preparation • Lysate preparation and hybridization • HPLC setup and analysis • HPLC data processing and data evaluation • Disposal of study samples	03, 05, 11 & 18-Sep-2024	TSM: 11-Oct-2024 SD: 15-Oct-2024
Audit of Report and Data	Draft Report Version from 05-Dec-2024	12-Dec-2024	SD: 12-Dec-2024 TSM: 17-Dec-2024

TSM: Test Site Management. SD: Study Director.

In addition to the listed process-based inspections, facility-based inspections take place yearly.

Quality Assurance Axolabs:	Cornelia Schumann	17.Dec.2024
	Name	Date and Signature

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LIST OF ABBREVIATIONS

AB0X	Analysis Batch, AN405.0-24_01_AB0X
ACN	Acetonitrile
AEX-HPLC	Anion-Exchange High-Performance Liquid Chromatography
BQL	Below Quantification Limit (Below the LLOQ)
CHMP	Committee for Human Medicinal Products
CoA	Certificate of Analysis
Conc.	Concentration
CSF	Cerebrospinal fluoid
CV	Coefficient of Variation
DPBS	Dulbecco´s Phosphate Buffered Saline
DNA	Deoxyribonucleic Acid
EDTA	Ethylenediaminetetraacetic Acid
EMA	European Medicines Agency
EU	European Union
FTS	Freeze-Thaw Stability
GCP	Good Clinical Practice
GLP	Good Laboratory Practice
ICH	International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use
LTS	Long-Term Stability
LLOQ	Lower Limit of Quantification
MDR	Method Development Report
n.a.	Not Applicable
OECD	Organization for Economic Co-operation and Development
PNA	Peptide Nucleic Acid
PrP	Prion Protein
PSS	Processed Sample Stability
QA	Quality Assurance
QC	Quality Control
SD	Standard Deviation
siRNA	small interfering Ribonucleic Acid
SOP	Standard Operating Procedure
SST	System Suitability Test(ing)
STM	Standard Test Method
STS	Short-Term Stability
ULOQ	Upper Limit of Quantification
UV	Ultraviolet

1 SUMMARY

This report summarizes the results of the bioanalytical method validation for the analysis of 2439-exNA in rat CSF (Axolabs Project: AN405.0-24).

The assay is based on the specific hybridization of the antisense strands of 2439-exNA with a complementary peptide nucleic acid (PNA) strand in a 1:1 stoichiometry. The PNA is labeled with an Atto425 fluorescence dye and yields a specific signal in the subsequent analysis by Peptide Nucleic Acid Anion-Exchange High-Performance Liquid Chromatography (PNA-AEX-HPLC) coupled to a fluorescence detector.

The method was developed without the need for any analyte extraction from rat CSF. Therefore, recovery of 100 % is maintained over the process (no internal standard is needed).

The bioanalytical method was previously developed at Axolabs GmbH and is described in the Axolabs Method Development Report (MDR) *MDR_2439exNA_CSF* (edition 01, effective date: 21-Aug-2024) that was used as the basis for this method validation (Supplemental File 1). During the method development phase, the suitability of the method for rat CSF was shown.

The following parameters described in the MDR had to be adjusted/adapted in the method validation:

- In the method development a lyophilized reference standard was used, which was weighed and dissolved. In the validation, an already prepared reference standard solution was used, which was provided by the Sponsor.
- Selectivity and matrix effect testing were performed using only half amount of the sample volume. Due to the small amount of available rat CSF, the sample volume had to be halved for these tests to save biological matrix. Since the results for these tests were all within the acceptance criteria, it is also possible to prepare and analyze only half of the sample volume for this method.

During the method development, two special requirements for this method were identified that were important for method validation.

The first requirement is to use artificial CSF as surrogate matrix. Sprague-Dawley rat CSF was not used as blank matrix in the method development, because it is a rare matrix and difficult to purchase. This procedure is in accordance with the ICH M10 guideline and the 3Rs Principle (Replacement, Reduction and Refinement) intended to reduce animal experiments to an absolute minimum. The composition of the artificial CSF is chosen to be as similar as possible to the biological CSF. In terms of animal welfare, artificial CSF is therefore the best alternative to replace biological matrix. In addition, the suitability of the artificial CSF was demonstrated during this method validation as part of selectivity and matrix effect testing – those tests were performed with the Sprague-Dawley rat CSF, while the calibration samples were prepared with the surrogate matrix. For all other validation tests, artificial CSF was used.

The second requirement is to prevent non-specific binding effects of the analyte 2439-exNA in CSF by adding a solution of Tween 20 and TCEP. Nonspecific binding is a problem in bioanalytical assays where a compound can adsorb to the container surface, resulting in a

reduced recovery of the analyte during analysis. The adsorption happens frequently in CSF because this matrix lacks proteins and lipids that can bind to the analytes or solubilize lipophilic analytes.

This problem also effects the preparation of stability samples. There, the stock solution of the analyte was diluted to obtain the working solutions for QC-Low, QC-High and QC-500xULOQ, which are then spiked into matrix. Due to the non-specific binding effects to pipette tips and the surface of vials during dilution series, it was not possible to spike the desired amount into the stability samples. To prevent unspecific binding, artificial CSF and a reagent (1 % Tween 20 / 1 mM TCEP diluted with artificial CSF) was used for the dilution of working solutions. The preparation of the stability samples was slightly deviating from the standard procedure described in the *ICH guideline M10 on bioanalytical method validation and study sample analysis* (EMA/CHMP/ICH/172948/2019; European Medicines Agency; 21 January 2023) (see Section 9.10).
To avoid possible adhesion effects in study samples, the planned procedure for sample processing is to add 5 % (v/v) of an 8 % Tween 20 / 50 mM TCEP solution directly to the study samples after thawing and before any pipetting step. This procedure was reflected in the stability testing of this validation (see Section 9.10). As this 8 % Tween 20 / 50 mM TCEP solution was also added to all calibration standard and QC samples in the same ratio as in study or LTS samples, no formal dilution of the study/LTS samples had to be calculated.

The final conditions of the validated method are described in the Standard Test Method (STM) *STM_Broad_2439-exNA_RatCSF* (revision 01, effective date: 04-Oct-2024; Supplemental File 2).

Additional bioanalytical method validations for 2439-exNA conducted at Axolabs are listed in Table 2.

Table 2: Summary Information of additional Method Validations for 2439-exNA

Axolabs Project	Type of Assay	Details of Validation	Date of Final Report
AN402.0-24	PNA-HPLC-Assay with Fluorescence Detection	other species and matrix	04-Dec-2024
AN403.0-24	PNA-HPLC-Assay with Fluorescence Detection	other matrix	03-Dec-2024

The work was done according to the Method Validation Plan signed on 12-Sep-2024 and Amendment 01 signed on 05-Dec-2024 by the Sponsor's Representative.

The results of the method validation are summarized in Table 3.

Table 3: Overall Summary Table

Items	Results	Section/Table	Comments
Analyte	2439-exNA	Section 3	n.a.
Reference Standard	2439-exNA	Section 4	n.a.
Methodology	PNA-AEX-HPLC-Assay with Fluorescence Detection	Section 8.1	n.a.
Biological Matrix	Rat CSF	Section 5 Table 4	n.a.
Calibration Curve Range	1.0 ng/mL – 1,000.0 ng/mL	Section 9.2 Table 10	Test was performed using artificial CSF
Selectivity	5 times higher response (peak area) in the spiked QC-LLOQ samples compared to the signal in the blank samples	Section 9.3 Table 12	Test was performed using 3 different Sprague-Dawley rat CSF lots
Matrix Effect	No matrix effect observed for 3 individually tested matrix lots	Section 9.4 Table 13	Test was performed using 3 different Sprague-Dawley rat CSF lots
Carry-Over	<20 % of the response (peak area) of the LLOQ calibration standard.	Section 9.5 Table 14	Test was performed using artificial CSF
Between-Run Accuracy	QC-LLOQ (1.0 ng/mL): 91.7 % QC-Low (3.0 ng/mL): 97.8 % QC-Mid (450.0 ng/mL): 102.7 % QC-High (900.0 ng/mL): 102.1 %	Section 9.7.2 Table 15	Test was performed using artificial CSF
Between-Run Precision	QC-LLOQ (1.0 ng/mL): 12.6 % QC-Low (3.0 ng/mL): 3.5 % QC-Mid (450.0 ng/mL): 1.4 % QC-High (900.0 ng/mL): 1.1 %	Section 9.8.2 Table 15	Test was performed using artificial CSF
Dilution Integrity	QC-500xULOQ : 500,000.0 ng/mL (Dilution factor: 1,000) Accuracy: 104.2 % Precision: 1.4 %	Section 9.9 Table 16	Test was performed using artificial CSF
Long-Term Stability	Ongoing (T= 0) at -80 °C ± 20 °C	Section 9.10.1 Table 17	Test was performed using artificial CSF
Freeze-Thaw and Short-Term Stability	Demonstrated for 5 cycles at -80 °C ± 20 °C and for 4 hours on ice	Section 9.10.2 Table 18	
Bench-Top Stability (Robustness)	Demonstrated 3 hours at room temperature after 1:1,000 dilution	Section 9.10.3 Table 19	Test was performed using artificial CSF
Processed Sample Stability	Demonstrated for 79 hours at 20 °C ± 3 °C	Section 9.10.4 Table 20	Test was performed using artificial CSF
Short-Term Stability of Stock and Working Solution	Demonstrated for 4 hours at room temperature	Section 9.10.5.1 Table 21	n.a.
Long-Term Stability of Stock Solution	Was not performed as data were provided by the Sponsor.		

2 COMPLIANCE

The method validation was conducted in compliance with the following guidelines, regulations, or guidance documents:

- ICH guideline M10 on bioanalytical method validation and study sample analysis (EMA/CHMP/ICH/172948/2019); European Medicines Agency; 21 January 2023

Additionally, this method validation project was conducted following the rules as depicted in the *OECD Principles of Good Laboratory Practice* (Revision from 1997).

3 ANALYTE

The analyte 2439-exNA is a prion protein (PrP)-lowering divalent siRNA (Figure 1). The sense strands of the two identical siRNAs are linked to each other via a tetraethylene glycol linker.

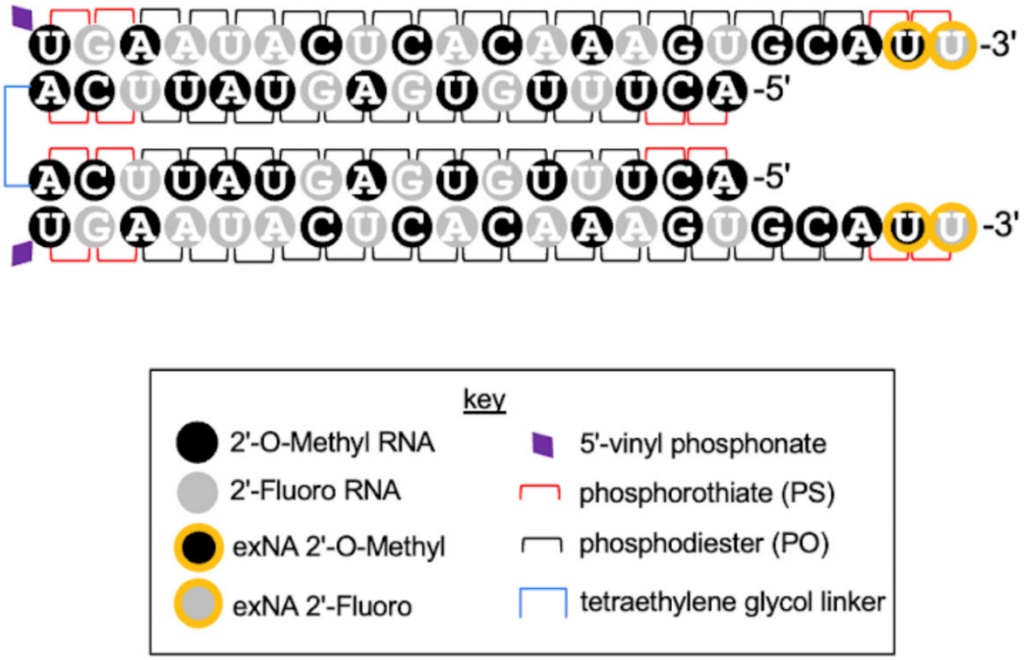


Figure 1: Structure of the PrP-lowering divalent siRNA 2439-exNA

4 Reference Standard

Identification:	PrP-lowering divalent siRNA 2439-exNA (shortened to 2439-exNA in this document)
Structure:	Refer to Section 3
Physical Description:	Slightly yellow solution
Purity:	96.6 % (determined by non-denaturing HPLC)
Total Oligo Concentration:	9.9 mg/mL (determined by UV spectrophotometry)
Full-Length Concentration:	9.5634 mg/mL (corrected for purity stated on the CoA)
Supplier:	Argonaut Manufacturing Services (Carlsbad, CA, USA)
Lot No.:	240621005
Manufacturing Date:	24-Jun-2024
Retest Date:	24-Jun-2026

With the exception of a document-based inspection and control of the corresponding Certificate of Analysis (CoA), no further testing on identity and purity of the reference standard was conducted upon receipt.

As the reference standard was supplied by Argonaut Manufacturing Services as a liquid formulation and could be used directly for the preparation of the working solutions (using a solution of 1 % Tween 20 and 1 mM TCEP in artificial CSF as diluent), separate aliquots were used for the calibration standards and the Quality Control (QC) samples. For this purpose, the reference standard was previously aliquoted into 100µL aliquots (Eppendorf Protein LoBind tubes). After aliquoting, the reference standard was renamed in accordance with internal nomenclature.

The concentrations of calibration standards and QC samples refer to the full-length concentration of 2439-exNA.

5 Blank Matrices

Blank matrices used in this method validation are listed in Table 4.

Table 4: Blank Matrices

Blank Matrix	Sex	Receipt Date	Supplier	Lot No.	Expiry Date	Analysis Batch
Sprague-Dawley rat CSF	n.a.	28-May-2024	BioIVT Elevating Science, Burgess Hill, UK	RAT572673_28May2024	28-May-2029	AB01
	n.a.			RAT572674_28May2024		
	n.a.			RAT572675_28May2024		
Artificial CSF	n.a.	12-Aug-2024	VWR International GmbH, Darmstadt, Germany	240719	19-Jul-2025	AB01 – AB03

6 PNA Probe

PNA Probe: PNA-BroadInst-01 from HLB Panagene Co., Ltd. (Daejeon, South Korea):

- N-terminus-Atto425 – OO – cdc ttt gtg dgt d – C-terminus, complementary to the antisense strands of 2439-exNA (OO: polyethylene glycol linker; d: 2,6-diaminopurine base modification)
  - Bulk powder was dissolved in 10 % acetonitrile (ACN) and 0.01 % Tween 20
  - Stock solution was diluted with 10 % ACN and 0.01 % Tween 20 to 2 µM (working solution)
- Lot No.: 241164
- Expiry Date: 22-Jul-2027

7 Reference Standard, PNA Probe, and Blank Matrices – Inventory and Disposition

Records of the receipt, distribution and storage of the reference standard(s), PNA probe(s), and blank matrices were maintained.

Unused reference standard(s) will be retained for use in subsequent studies for the Sponsor.

Unused PNA probe(s) and blank matrices will be retained for use in subsequent studies.

Expired reference standard(s) will be disposed of unless the shelf life can be extended (new CoA required).

Expired PNA probe(s) and blank matrices will be disposed of or transferred to non-GLP storage.

8 METHOD

8.1 Method Summary

This report describes the validation of an AEX-HPLC method with fluorescence detection that allows the sensitive and specific detection of the antisense strands of 2439-exNA, a prion protein (PrP)-lowering divalent siRNA, in rat CSF samples.

The assay is based on the specific hybridization of a 13-mer PNA probe conjugated at the N-terminus with an Atto425 dye that is complementary to the antisense strands of the divalent siRNA 2439-exNA.

The assay is a one-pot approach in which rat CSF lysates are prepared by Proteinase K treatment and the antisense strands of 2439-exNA are hybridized with the PNA in the same step. Subsequently, the samples are analyzed by AEX-HPLC coupled to a fluorescence detector.

Contrary to standard procedure, the addition of 5 % (v/v) of an 8 % Tween 20 and 50 mM TCEP solution directly to the study samples after thawing and before any pipetting step is necessary to prevent nonspecific binding of the analyte to the container surface. Besides that, the working solutions and samples are diluted with 1 % Tween 20 and 1 mM TCEP in artificial CSF. Also, the injected sample volume is 50 µL, resulting in 2.5 µL rat / artificial CSF on column.

Quantification was performed based on an external calibration curve generated from a standard dilution series in artificial CSF. Linear calibration curves (weighted 1/X) were calculated from 1.0 ng/mL to 1,000.0 ng/mL in artificial CSF. The concentrations of the calibration standards refer to the concentration of full-length 2439-exNA.

The detailed procedure is described in the STM, which is included as a supplemental file to this report.

8.2 Applicable Documents

- Axolabs Method Validation Plan and Amendment 01 thereto for the analysis of 2439-exNA in rat CSF (Axolabs Project: AN405.0-24)
- Method Development Report (MDR) *MDR_2439exNA_CSF* (edition: 01, effective date: 21-Aug-2024)
- Standard Test Method (STM) *STM_Broad_2439-exNA_RatCSF* (revision 01, effective date: 04-Oct-2024)
- KU-RGL-SOP-002, Revision 03 (formerly SOP REG-102) – Methoden-Validierung im GLP/GCP-Bereich (*Method Validation in GLP/GCP Area*)
- KU-RGL-SOP-009, Revision 03 (formerly SOP REG-109) – Durchführung des PNA-HPLC-Assay (*Execution of the PNA-HPLC-Assay*)
  - KU-RGL-INS-005, Revision 05 (formerly REG-109-I01) – PNA-HPLC-Assay SST

8.3 Computerized Systems

Critical computerized systems used in this method validation are listed in Table 5.

Table 5: Computerized Systems

System Name	Revision / Version	System Description
Thermo Fisher Ultimate 3000/Shimadzu LC-2050C HPLC System with a Shimadzu RF-20Axs Fluorescence Detector used with Chromeleon HPLC Software	7.2 SR5	HPLC Data collection and export to Excel
Microsoft Excel Spreadsheet: KU-RGL-ESS-002 (formerly REG-102-S02) – Method Validation Data Report	03	Calculates results of calibration standard and QC sample accuracy and precision
Microsoft Excel Spreadsheet: KU-RGL-ESS-005 (formerly REG-102-S05 – Method Validation Protocol PNA Assay - Liquid Matrices - OnePot	01	Used as instruction and as the protocol for the assay execution for liquid matrices using the OnePot method: Calculates all sample preparation steps and the HPLC sequence for an analysis batch
Microsoft Excel Spreadsheet: KU-RGL-ESS-014 (formerly REG-109-S04 – Preparation of HPLC Buffers & Stock Solutions	02	Calculates all buffer preparation steps, including the preparation of the required stock solutions

8.4 HPLC System and Parameters

The HPLC systems used for the AEX-HPLC analyses are listed in Table 6.

Table 6: HPLC Systems

HPLC System	Module	Supplier	Model	Serial No.
HPLC-GLP-7 (used for AB03)	Degasser/Solvent Rack	Thermo Fisher Scientific	Ultimate SR-3000	7294808
	Pump		Ultimate LPG-3400SD	8135975
	Autosampler		Ultimate WPS-3000TSL	8135990
	Column Oven		Ultimate TCC-3000RS	6016238
	Fluorescence Detector	Shimadzu	RF-20AXS	L20505573772
	Communication Unit		CBM-20A	L20235570188
HPLC System	**Module**	**Supplier**	**Model**	**Serial No.**
HPLC-GLP-S-18 (used for AB01 and AB02)	HPLC Kompakt System	Shimadzu	LC-2050C	L22975950036
	Fluorescence Detector		RF-20AXS	L20505974379

- Eluent Buffer A: 25 mM Trizma hydrochloride solution (pH 8), 1 mM EDTA, 30 % ACN, 3 M urea
- Eluent Buffer B: 25 mM Trizma hydrochloride solution (pH 8), 1 mM EDTA, 30 % ACN, 3 M urea, 1 M NaBr
- Eluent Buffer C: 10 % ACN, 4 M NaClO4, 2 M urea, 0.5x triphosphate buffer, adjusted to pH 2.5 with phosphoric acid
- Flow Rate: 1 mL/min
- Detection: Excitation @ 436 nm; Emission @ 484 nm; Gain: Medium 16
- Injection of 50 µL of the hybridized sample (corresponds to 2.5 µL matrix on column)
- Column Temperature: 45 °C
- Autosampler Temperature: 20 °C
- HPLC Program: Broad

Table 7: HPLC Gradient Table

Time [min]	% Eluent A	% Eluent B	% Eluent C	Curve
0.00	82	18	0	linear
1.00	82	18	0	linear
5.00	50	50	0	linear
5.10	0	0	100	linear
6.10	0	0	100	linear
6.20	82	18	0	linear
6.70	82	18	0	linear
6.80	0	0	100	linear
7.80	0	0	100	linear
7.90	82	18	0	linear
8.40	82	18	0	linear
8.50	0	0	100	linear
9.50	0	0	100	linear
9.60	82	18	0	linear
13.50	82	18	0	linear

8.5 HPLC Columns

Thermo Fisher Scientific DNAPac PA200 (4 x 250 mm) Column; Serial Number 010629; Lot Number 02306174 (used for AB01 and AB02)

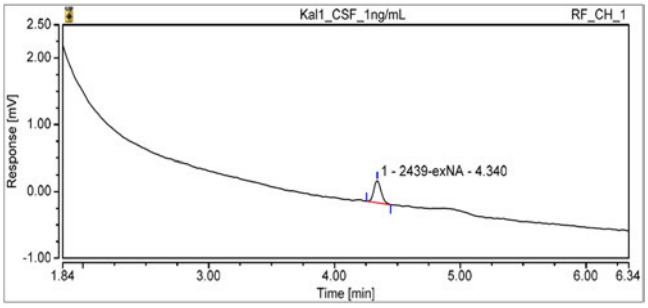
Thermo Fisher Scientific DNAPac PA200 (4 x 250 mm) Column; Serial Number 010628; Lot Number 02306174 (used for AB03)

8.6 Data Evaluation

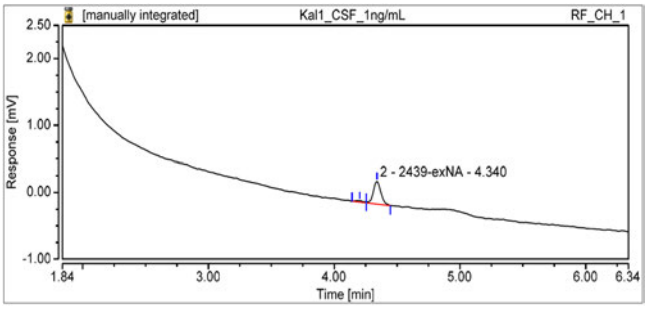
Data acquisition and integration were performed using the software package Chromeleon (Version 7.2) from Thermo Fisher Scientific.

For every injection, the peaks were integrated automatically by the HPLC chromatography software but are adjusted manually if the baseline and peak shoulders are not detected correctly by the HPLC software. When necessary, manual integration was performed by vertical peak splits at minima between two not fully resolved peaks. General rules for manual integration are documented in KU-RGL-SOP-009 (formerly SOP REG-109).

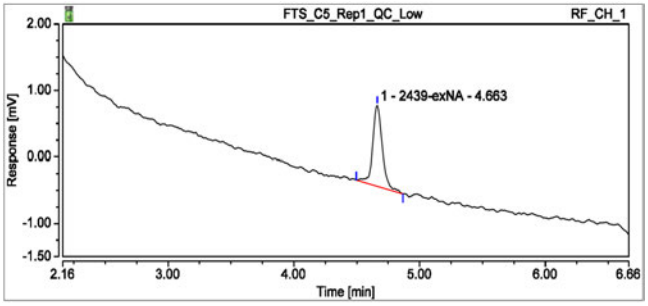
A report format of this file is generated containing values of retention time, peak area, and concentration of 2439-exNA in rat / artificial CSF and was saved in PDF format.



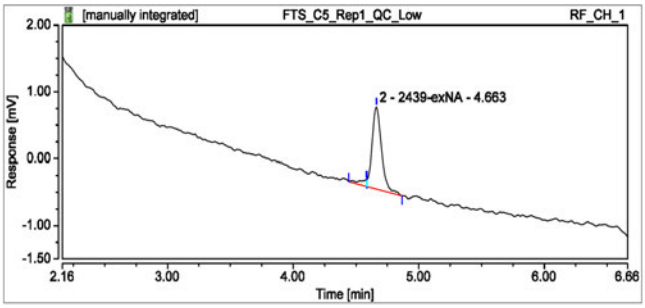
HPLC chromatogram of Cal-level 2 – 1.0 ng/mL, sample no. 4 in AB01 **(original)**



HPLC chromatogram of Cal-level 2 – 1.0 ng/mL, sample no. 4 in AB01 **(re-integrated)**



HPLC chromatogram of QC-Low FTS sample (5 cycles), sample no. 40 in AB03 **(original)**



HPLC chromatogram of QC-Low FTS sample (5 cycles), sample no. 40 in AB03 **(re-integrated)**

Figure 2: Representative HPLC chromatograms (original and re-integrated)

8.7 Statistical Analysis

Mean values, standard deviations (SD), and coefficients of variation (CV) were calculated using validated Microsoft Excel spreadsheets.

9 Results and Discussion

9.1 System Suitability Test (SST)

The reproducibility of the HPLC system was determined by conducting an SST according to KU-RGL-INS-005 (formerly REG-109-I01) before measuring an analysis batch. Acceptance criteria are described in KU-RGL-INS-005 (formerly REG-109-I01), and the results of the SST run are attached to the documentation of each analysis batch.

A successful SST was completed before each reported analysis batch.

9.2 Calibration Curve

Calibration standards were generated by spiking known concentrations of 2439-exNA into artificial CSF. The linear range was determined from 1.0 ng/mL to 1,000.0 ng/mL. The

calibration levels used were 1.0 ng/mL, 5.0 ng/mL, 50.0 ng/mL, 100.0 ng/mL, 500.0 ng/mL, 1,000.0 ng/mL and one blank sample (artificial CSF processed in the absence of 2439-exNA).

The calibration curves were prepared in duplicate for each analysis batch. Blank samples and calibration standards not meeting the acceptance criteria were not used to calculate the calibration curves.

In total 6 calibration curves were analyzed in 3 different analysis batches.

9.2.1 Concentration-Response

The concentration-response relation was calculated by linear regression of the mean peak responses of the 6 calibration standards using 1/X weighting.

The calibration curve parameters (slope, y-axis offset, and coefficient of determination) from reported analysis batches are listed in Table 9.

9.2.2 Calibration Curve Accuracy

Acceptance Criteria:

The back-calculated concentrations of the calibration standards should be within ± 15 % of the nominal concentrations (± 20 % for the LLOQ calibration standard).

At least 75 % of the calibration standards, with a minimum of 6 calibration standard levels, and at least 50 % of the calibration standards per concentration level must fulfill this criterion.

Results:

- 100.0 % of the back-calculated concentrations for all calibration standards were within ± 15 % (± 20 % for the LLOQ calibration standard) of the nominal concentrations.

Data are listed in Table 10.

Based on these results, the method can be used to determine concentrations of 2439-exNA in artificial / rat CSF over the concentration range of 1.0 ng/mL to 1,000.0 ng/mL.

9.2.3 Sensitivity/Lower Limit of Quantification (LLOQ)

The sensitivity of the assay is defined as the lowest 2439-exNA concentration that can be quantified with acceptable accuracy and precision – i.e., the LLOQ.

To confirm the LLOQ, 6 individually prepared samples with a nominal concentration at the LLOQ were analyzed (LLOQ calibration standards).

Acceptance Criteria:

The analyte response (peak area) in each LLOQ calibration standard should be at least 5 times higher than the response (peak area) at the retention time of the analyte in the blank sample of the same calibration curve.

The mean back-calculated concentration of the 6 LLOQ calibration standards should be within ± 20 % of the nominal concentration and the coefficient of variation (CV) should not exceed 20 %.

Results:

- In 2 of 6 blank samples, a signal for 2439-exNA was detected, but 2439-exNA response (peak area) in each LLOQ calibration standard was at least 5 times higher than the response (peak area) in the blank sample of the same calibration curve.
- The mean back-calculated concentration of the 6 LLOQ calibration standards was 96.7 % of the nominal concentration with a CV of 5.3 %.

Data are listed in Table 10 and Table 11.

The LLOQ for 2439-exNA was confirmed at 1.0 ng/mL in artificial CSF. The LLOQ can also be applied when using rat CSF.

9.2.4 Upper Limit of Quantification (ULOQ)

The highest calibration standard in the calibration curve defines the ULOQ.

Six (6) individually prepared samples with a nominal concentration at the ULOQ were analyzed (ULOQ calibration standards).

Acceptance Criteria:

The mean back-calculated concentration of the 6 ULOQ calibration standards should be within ± 15 % of the nominal concentration and the CV should not exceed 15 %.

Results:

- The mean back-calculated concentration of the 6 ULOQ calibration standards was 99.7 % of the nominal concentration with a CV of 1.2 %.

Data are listed in Table 10.

The ULOQ for 2439-exNA was confirmed at 1,000.0 ng/mL in artificial CSF. The ULOQ can also be applied when using rat CSF.

9.3 Selectivity

The selectivity of the method was evaluated by comparison of unspiked samples of 3 different blank rat CSF lots with the same rat CSF samples spiked with 2439-exNA at the concentration of the LLOQ (QC-LLOQ).

Acceptance Criteria:

The analyte response (peak area) in each QC-LLOQ sample should be at least 5 times higher than the response (peak area) at the retention time of the analyte in the blank sample of the same matrix lot.

The back-calculated concentration each QC-LLOQ sample should be within ± 20 % of the nominal concentration.

Results:

- In 2 of 3 selectivity blank samples, a signal at the retention time of 2439-exNA was detected, but the 2439-exNA response (peak area) in each QC-LLOQ sample was at least 5 times higher than the response (peak area) in the blank sample of the same matrix lot.
- The back-calculated concentrations of the QC-LLOQ samples ranged from 80.0 % – 90.0 % of the nominal concentration.

Data are listed in Table 12.

Selectivity of the method was confirmed for rat CSF.

9.4 Matrix Effect

The matrix effect was evaluated by analyzing triplicates of QC-Low and QC-High samples, each prepared using 3 different blank rat CSF lots.

Acceptance Criteria:

For each evaluated blank rat CSF lot, the mean back-calculated concentration of the QC-Low and QC-High samples (each triplicates) should be within ± 15 % of the nominal concentrations and the CV should not exceed 15 %.

Results:

- For 3 matrix lots (RAT572673_28May2024, RAT572674_28May2024, RAT572675_28May2024), the mean back-calculated concentrations of the QC samples ranged from 101.1 % – 104.4 % (QC-Low) and 101.8 % – 102.0 % (QC-High) of the nominal concentrations, and CV values ranged from 1.9 % – 4.9 % (QC-Low) and 0.4 % – 0.5 % (QC-High).

Data are listed in Table 13.

Based on these results, there is no matrix effect for the assay regarding the analysis of 2439-exNA in rat CSF.

9.5 Carry-Over

Blank samples were injected after the ULOQ calibration standards to assess carry-over.

Acceptance Criteria:

The analyte response (peak area) in each LLOQ calibration standard should be at least 5 times higher than the analyte response (peak area) in the carry-over blank sample following the ULOQ calibration standard of the same calibration curve.

Results:

- In 1 of 6 carry-over samples, a signal for 2439-exNA was detected, but 2439-exNA response (peak area) in each LLOQ calibration standard was at least 5 times higher than the response (peak area) in the carry-over blank sample following the ULOQ calibration standard of the same calibration curve.

Data are listed in Table 14.

No carry-over was observed for 2439-exNA in artificial CSF. The result can also be applied when using rat CSF. Therefore, no special precautions after the analysis of samples with a high concentration of 2439-exNA have to be implemented.

9.6 Quality Control (QC) Samples

QC samples were prepared by spiking known amounts of 2439-exNA into artificial CSF.

QC samples were prepared at 4 concentration levels which cover the calibration curve range:

- at the LLOQ (QC-LLOQ): 1.0 ng/mL
- within 3 times above the LLOQ (QC-Low): 3.0 ng/mL
- at 30 – 50 % of the calibration curve range (QC-Mid): 450.0 ng/mL
- at least at 75 % of the upper calibration curve range (QC-High): 900.0 ng/mL

For evaluation of dilution integrity, freeze-thaw, long-term and short-term stability and robustness an additional QC concentration level was prepared:

- at 500 times above the ULOQ (QC-500xULOQ): 500,000.0 ng/mL

9.7 Accuracy

The accuracy of the analytical method describes the closeness of the determined concentration obtained by the method to the nominal concentration of 2439-exNA (expressed in percentage).

The analysis of assay accuracy was conducted using the QC samples at the 4 concentration levels that were measured as 6 individually prepared replicates per analytical run.

The analysis of assay accuracy was divided into within-run and between-run accuracy. The within-run accuracy was repeated 3 times, prepared by 2 technicians on different days, and analyzed on 2 independent HPLC systems. Thus, 3 independent data sets for the between-run accuracy were generated.

The concentration of 2439-exNA was calculated against the external calibration curve, generated in duplicate for each data set.

9.7.1 Within-Run Accuracy

Acceptance Criteria:

The mean back-calculated concentration of the QC samples at each of the 4 concentration levels and in each of the 3 data sets should be within ± 15 % of the nominal concentrations (± 20 % for the QC-LLOQ).

Results:

- The mean back-calculated concentrations of the QC samples at each of the 4 concentration levels in the 3 data sets ranged from 83.3 % – 105.0 % (QC-LLOQ), 95.6 % – 101.7 % (QC-Low), 102.0 % – 103.8 % (QC-Mid) and 101.5 % – 102.7 % (QC-High) of the nominal concentrations.

Data are listed in Table 15.

9.7.2 Between-Run Accuracy

Acceptance Criteria:

The mean back-calculated concentrations of all QC samples at each of the 4 concentration levels of all 3 data sets should be within ± 15 % of the nominal concentrations (± 20 % for the QC-LLOQ).

Results:

- The mean back-calculated concentrations of all QC samples at each of 4 concentration levels of all 3 data sets were 91.7 % (QC-LLOQ), 97.8 % (QC-Low), 102.7 % (QC-Mid), and 102.1 % (QC-High) of the nominal concentrations.

Data are listed in Table 15.

Based on these results, the assay for 2439-exNA in artificial CSF is considered accurate. The result can also be applied when using rat CSF.

9.8 Precision

The precision of the analytical method describes the closeness of repeated individual measures of the 2439-exNA concentration, expressed as coefficient of variation (CV).

The analysis of assay precision was conducted using the QC samples at the 4 concentration levels. All concentration levels were measured as 6 individually prepared replicates per analytical run.

As for accuracy, precision was demonstrated within a single analytical run and between different analytical runs. The same analytical runs and data were used for the demonstration of accuracy.

The concentration of 2439-exNA was calculated against the external calibration curve, generated in duplicate for each data set.

9.8.1 Within-Run Precision

Acceptance Criteria:

The within-run CV of the QC samples at each of the 4 concentration levels and in each of the 3 data sets should not exceed 15 % (20 % for the QC-LLOQ).

Results:

- The within-run CV of the QC samples at each of the 4 concentration levels in the 3 data sets ranged from 6.0 % – 8.0 % (QC-LLOQ), 1.8 % – 2.6 % (QC-Low), 0.4 % – 2.0 % (QC-Mid) and 0.3 % – 1.7 % (QC-High).

Data are listed in Table 15.

9.8.2 Between-Run Precision

Between-run precision was determined with an identical setup as for within-run precision but included the assessment of QC samples from 3 independent data sets prepared by 2 technicians on 3 different days on 2 independent HPLC systems.

Acceptance Criteria:

The between-run CV of all QC samples at each of the 4 concentration levels of all 3 data sets should not exceed 15 % (20 % for the QC-LLOQ).

Results:

- The between-run CV of all QC samples at each of the 4 concentration levels of all 3 data sets were 12.6 % (QC-LLOQ), 3.5 % (QC-Low), 1.4 % (QC-Mid), and 1.1 % (QC-High).

Data are listed in Table 15.

Based on these results, the assay for 2439-exNA in artificial CSF is considered precise. The result can also be applied when using rat CSF.

9.9 Dilution Integrity

Dilution integrity was evaluated using QC samples prepared at the concentration level 500-fold (500,000.0 ng/mL) above the ULOQ (6 replicates for each dilution factor).

The 500,000.0 ng/mL QC samples (QC-500xULOQ) were analyzed and quantified after a 1:1,000 dilution.

The QC samples were diluted with 1 % Tween 20 and 1 mM TCEP in artificial CSF. During the method development, the suitability of artificial CSF / 1 % Tween 20 / 1 mM TCEP as a diluent for study samples was shown. Blank rat CSF was not used for sample dilution since rat CSF is a rare matrix. In line with the 3R principles (Reduce, Refine, Replace) it is not ethically justifiable to use the large quantities of blank rat CSF for sample dilution. In terms of animal welfare, artificial CSF is therefore a good alternative to replace biological matrix. The addition of 1 % Tween 20 / 1 mM TCEP to artificial CSF for the dilution of the samples is necessary to avoid non-specific binding problems.

Acceptance Criteria

For each evaluated dilution factor and concentration, the mean back-calculated concentration of the QC samples should be within ± 15 % of the nominal concentration and the CV should not exceed 15 %.

Results:

- The mean back-calculated concentration was 104.2 % for the 500,000.0 ng/mL QC samples (1:1,000 dilution).
- The CV was 1.4 % for the 500,000.0 ng/mL QC samples (1:1,000 dilution).

Data are listed in Table 16.

Dilution of samples up to 1:1,000 up to a concentration of 500,000.0 ng/mL met the acceptance criteria for precision and accuracy. As the 1:1,000 dilution integrity testing fulfilled the acceptance criteria, no additional dilutions (e.g. 1:500) were tested.

9.10 Stability

The stability of 2439-exNA was analyzed under several conditions relevant to the assay setup. 2439-exNA was spiked at 3 concentration levels into artificial CSF samples and analyzed in triplicate. The concentration levels were identical to the QC-Low, QC-High and QC-500xULOQ samples, with concentrations of 3.0 ng/mL (QC-Low), 900.0 ng/mL (QC-High) and 500,000.0 ng/mL (QC-500xULOQ). The QC-500xULOQ samples were evaluated after 1:1,000

dilution (using 1 % Tween 20 and 1 mM TCEP in artificial CSF) to mimic potential samples above the validated ULOQ. Calibration curves were freshly prepared for each analytical run.

In order to prevent non-specific adhesive effects when preparing the stability samples, the working solutions of 2439-exNA were prepared with 1 % Tween 20 and 1 mM TCEP in artificial CSF. To mimic the planned procedure for the CSF study samples, 5 % (v/v) of a 8 % Tween 20 and 50 mM TCEP solution was added directly to the LTS sample vials after first thawing and before pipetting. As this was also added to calibration standard and QC samples in the digestion step, no formal dilution of study samples occurs.

As no pre-dilutions are necessary for preparing these samples, no Tween 20 / TCEP will be added to the samples before the first thawing. After thawing and before pipetting the sample, 5 % (v/v) of a 8 % Tween 20 and 50 mM TCEP solution was added.

Long-term stability, freeze-thaw stability, and short-term stability were evaluated in polypropylene tubes (Protein LoBind Tubes).

To prevent non-specific adhesion effects, e.g. on the pipette tip during preparation of the stability samples, the stability samples were prepared slightly deviating from *ICH guideline M10 on bioanalytical method validation and study sample analysis* (EMA/CHMP/ICH/172948/2019; European Medicines Agency; 21 January 2023). The stability samples were prepared by first adding the matrix in each stability aliquot and then spiking the working solution, containing Tween 20 and TCEP, of each concentration level individually into the stability aliquots. This reduces the putative adhesion effects during aliquoting, as the ratio of Tween 20 and TCEP during aliquoting is constant. In contrast to the preparation method for stability samples according to the ICH guideline M10. Here, the working solution of each concentration level should first be spiked into the matrix and then aliquoted. However, as this procedure reduces the ratio of Tween 20 and TCEP in the final solution, the putative adhesion effects could be increased during aliquoting. For this reason, the stability samples will be prepared slightly deviating from ICH guideline M10. The triplicate analysis was then performed from one aliquot.

9.10.1 Long-Term Stability

Aliquots of 2439-exNA samples in artificial CSF at the 3 QC concentration levels were stored at -80 °C ± 20 °C, and were and will be analyzed at up to 3 time points (T= 0, 3, and 12 months) in triplicate.

Acceptance Criteria:

For each evaluated condition, the mean back-calculated concentration of the QC-Low, QC-High and QC-500xULOQ samples (each triplicates) should be within ± 15 % of the nominal concentrations.

Results:

At the time the Method Validation Report was prepared, results of the first time point (T= 0) were available.

Storage without 8 % Tween 20 / 50 mM TCEP solution in the original sample (addition of Tween 20 / TCEP solution after thawing):

- At T= 0 in AB01, the mean back-calculated concentrations were 93.3 % (QC-Low), 99.3 % (QC-High) and 99.4 % (QC-500xULOQ) of the nominal concentrations.

Storage with 8 % Tween 20 / 50 mM TCEP solution in the original sample (addition of Tween 20 / TCEP solution before freezing):

- At T= 0 in AB01, the mean back-calculated concentrations were 95.6 % (QC-Low), 96.9 % (QC-High) and 101.0 % (QC-500xULOQ) of the nominal concentrations.

Data are listed in Table 17.

The result can also be applied when using rat CSF.

Long-term stability of 2439-exNA in artificial CSF will be further monitored for up to 12 months. The final long-term stability data set will be added as an addendum to the Method Validation Report at the end of the study.

9.10.2 Freeze-Thaw and Short-Term Stability

Freeze-thaw and short-term stability of 2439-exNA in artificial CSF were evaluated in a combined test with added Tween 20 / TCEP.

Aliquots of 2439-exNA samples in artificial CSF at the 3 QC concentration levels were stored at -80 °C ± 20 °C for at least 12 hours.

Thawing of the QC samples from -80 °C ± 20 °C was done on ice. After thawing, the QC samples were stored on ice for 4 hours before further processing, covering short-term stability requirements.

Before refreezing the samples, 5 % of a (v/v) 8 % Tween 20 and 50 mM TCEP solution was added in order to mimic the procedure for study samples. Subsequently, the QC samples were refrozen for at least 12 hours at -80 °C ± 20 °C to start the next freeze-thaw cycle and to repeat the process described above, without addition of additional Tween 20 / TCEP solution. Like this, the first freeze-thaw/STS cycle reflects the study samples without added Tween 20 / TCEP and the following cycles reflect study samples after addition of Tween 20 / TCEP.

After 5 freeze-thaw cycles, the QC samples were processed and analyzed in triplicate according to the assay conditions.

Acceptance Criteria:

The mean back-calculated concentration of the QC-Low, QC-High and QC-500xULOQ samples (each triplicates) should be within ± 15 % of the nominal concentrations.

Results:

- After 5 freeze/thaw cycles with 4 hours on ice, the mean back-calculated concentrations were 106.7 % (QC-Low), 108.8 % (QC-High) and 100.1 % (QC-500xULOQ) of the nominal concentrations.

Data are listed in Table 18.

At the evaluated concentrations, 2439-exNA is stable in artificial CSF when thawed up to 5 times on ice during sample analysis and for up to 4 hours during sample storage on ice. The result can also be applied when using rat CSF.

9.10.3 Bench-Top Stability/Robustness

Since the assay is set up as a one-pot method that does not have separate lysis and hybridization steps, dilutions are the only reason for bench-top storage of study samples. Therefore, the robustness was tested with the QC-500xULOQ concentration, which was left at ambient conditions for 3 hours after a 1:1,000 dilution and prior to subsequent assay steps.

Acceptance Criteria:

The mean back-calculated concentration of the QC-500xULOQ samples (each triplicates) should be within ± 15 % of the nominal concentrations.

Results:

- After storage of 3 hours at ambient conditions after dilution and prior to the subsequent steps, the mean back-calculated concentration was 89.9 % (QC-500xULOQ) of the nominal concentration.

Data are listed in Table 19.

At the evaluated concentration, 2439-exNA in artificial CSF is stable during prolonged storage at ambient conditions for 3 hours after a 1:1,000 dilution and prior to subsequent assay steps. The result can also be applied when using rat CSF.

9.10.4 Stability of the Processed Sample/On-Instrument Stability (Autosampler)

Stability of the processed sample of 2439-exNA in artificial CSF was assessed with 4 QC sets of QC-Low and QC-High triplicates that were stored on the HPLC autosampler at 20 ± 3 °C and were analyzed at different time points, covering 79 hours between the analysis of the first and the last QC set.

Acceptance Criteria:

For each evaluated time point, the mean back-calculated concentration of the QC-Low and QC-High samples should be within ± 15 % of the nominal concentrations.

The QC set of last time point, which fulfills the acceptance criteria, defines processed sample stability.

Results:

- After time point 1, the mean back-calculated concentrations were 100.0 % (QC-Low) and 101.6 % (QC-High) of the nominal concentrations.
- After time point 2 (20 hours), the mean back-calculated concentrations were 102.2 % (QC-Low) and 102.3 % (QC-High) of the nominal concentrations.
- After time point 3 (46 hours), the mean back-calculated concentrations were 96.7 % (QC-Low) and 96.0 % (QC-High) of the nominal concentrations.
- After time point 4 (79 hours), the mean back-calculated concentrations were 96.7 % (QC-Low) and 96.1 % (QC-High) of the nominal concentrations.

Data are listed in Table 20.

At the evaluated concentrations, 2439-exNA in processed artificial CSF samples is stable during storage on the HPLC autosampler at 20 ± 3 °C for up to 79 hours. The result can also be applied when using rat CSF.

If a longer period of sample storage on the autosampler should be necessary, the last QC sample set and calibration curve will verify the validity of the analysis.

9.10.5 Stability of Stock Solution and Working Solutions

9.10.5.1 Short-Term Stability of Stock Solution and Working Solutions

Aliquots of the reference standard stock solution of 2439-exNA with a concentration of approximately 9.5634 mg/mL in 1x DPBS (total oligonucleotide concentration corrected for the purity stated in the CoA), and the highest and the lowest concentration of the working solutions (working solution LLOQ/ULOQ) of 2439-exNA were stored at room temperature for 4 hours and analyzed in triplicate. The reference standard stock solution was diluted to QC-High concentration before analysis.

The samples were analyzed by comparing the instrument's response with a freshly thawed reference standard stock solution and freshly prepared working solutions of the same concentrations.

Acceptance Criteria:

The mean back-calculated concentration of the QC-High samples (reference standard stock solution) and the working solution samples (each triplicates) should be within ± 15 % of the nominal concentrations (± 20 % for the working solution LLOQ).

Results:

After 4 hours at room temperature, the mean back-calculated concentrations were 90.0 % (working solution LLOQ), 99.0 % (working solution ULOQ), and 94.0 % (stock solution) of the nominal concentrations.

Data are listed in Table 21.

At the evaluated concentrations, 2439-exNA stock and working solutions covering the complete assay range are stable during storage for up to 4 hours at room temperature.

9.10.5.2 Long-Term Stability of Stock Solution

Long-term stability of the reference standard stock solution was not performed, as stability data are provided by the Sponsor.

10 Deviations

The bioanalytical method validation for the analysis of 2439-exNA in rat CSF was conducted according to the Method Validation Plan and to applicable SOPs without any deviation.

11 Conclusions

The bioanalytical method for the analysis of 2439-exNA in rat CSF was successfully validated and is described in the *STM_Broad_2439-exNA_RatCSF* (revision 01, effective date: 04-Oct-2024). All validation tests fulfilled the acceptance criteria described in the Method Validation Plan.

The method can be used within a linear range from 1.0 ng/mL to 1,000.0 ng/mL of 2439-exNA in artificial / rat CSF, applying a 1/X weighted linear regression model.

2439-exNA is stable in artificial / rat CSF during up to 5 freeze/thaw cycles. The evaluation of long-term stability of 2439-exNA in artificial CSF is still ongoing, and the results will be added as an addendum to this Method Validation Report after the finalization of the long-term stability testing.

12 List of Records to be Retained

All original data and documents generated during this method validation will be archived at Axolabs GmbH or at an Axolabs-employed GLP-compliant archive facility under appropriate

conditions in accordance with German and EU legislation, and KU-RGL-SOP-005 (formerly SOP REG-105).

The following records will be retained:

- Final signed Method Validation Plan and Amendment 01
- Critical communication, as appropriate
- Raw data:
  - HPLC file(s) with HPLC chromatograms, audit trails, and HPLC program
  - Assay protocols
  - System Suitability Test reports
  - CoA (or equivalent) of PNA probe, blank matrices, artificial CSF and reference standard
  - Other raw data (if applicable)
- QC and QA records of document and data reviews/audits
- Final signed Method Validation Report and all addenda

A backup file of all analytical HPLC data, together with an electronic PDF report of all analytical runs, and all electronic files generated during the method validation (reports, Excel-Files) are stored in electronic form.

13 DATA TABLES

Table 8: Analytical Batches for the Analysis of 2439-exNA in Artificial / Rat CSF

Run ID	Date of Analysis	Description	Status	Comments
AN405.0-24_01_AB01	18-Sep-2024	First Validationbatch	passed	n.a.
AN405.0-24_01_AB02	20-Sep-2024	Second Validationbatch	passed	n.a.
AN405.0-24_01_AB03	25-Sep-2024	Third Validationbatch	passed	n.a.

Table 9: Calibration Curve Parameters for the Analysis of 2439-exNA in Artificial CSF

Equation for the calibration curve using a linear regression model weighted 1/X: y = m*x + b					
Run ID	**Date of Analysis**	**Sample No.**	**Slope [m]**	**Y-Offset [b]**	**R-Squared**
AB01	18-Sep-2024	4 - 9 and 109 - 114	0.02481	0.00063	0.99992
AB02	20-Sep-2024	4 - 9 and 73 - 78	0.02669	0.00222	0.99988
AB03	25-Sep-2024	4 - 9 and 51 - 56	0.03325	-0.00277	0.99972

Table 10: Results of the Calibration Standards for the Analysis of 2439-exNA in Artificial CSF

		Cal-Level-2 (LLOQ)		Cal-Level-3		Cal-Level-4		Cal-Level-5		Cal-Level-6		Cal-Level-7 (ULOQ)	
		1.0 ng/mL		5.0 ng/mL		50.0 ng/mL		100.0 ng/mL		500.0 ng/mL		1,000.0 ng/mL	
Run ID	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 20 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**
AB01	4 - 9	0.9	90.0 %	4.9	98.0 %	50.2	100.4 %	101.0	101.0 %	495.2	99.0 %	989.4	98.9 %
AB01	109 – 114	1.0	100.0 %	5.1	102.0 %	51.4	102.8 %	105.1	105.1 %	503.0	100.6 %	1,004.8	100.5 %
AB02	4 – 9	1.0	100.0 %	5.0	100.0 %	51.4	102.8 %	101.9	101.9 %	496.1	99.2 %	983.7	98.4 %
AB02	73 – 78	0.9	90.0 %	5.0	100.0 %	52.3	104.6 %	105.1	105.1 %	503.9	100.8 %	1,005.8	100.6 %
AB03	4 – 9	1.0	100.0 %	4.9	98.0 %	48.4	96.8 %	101.4	101.4 %	480.0	96.0 %	987.6	98.8 %
AB03	51 – 56	1.0	100.0 %	5.0	100.0 %	52.2	104.4 %	103.0	103.0 %	515.6	103.1 %	1,011.8	101.2 %
	Mean	0.97	96.7 %	4.98	99.7 %	50.98	102.0 %	102.92	102.9 %	498.97	99.8 %	997.18	99.7 %
	SD (±)	0.05		0.08		1.47		1.82		11.83		11.66	
	CV (%)	5.3 %		1.5 %		2.9 %		1.8 %		2.4 %		1.2 %	
	N	6		6		6		6		6		6	

Table 11: Sensitivity/Lower Limit of Quantification (LLOQ) for the Analysis of 2439-exNA in Artificial CSF

		Cal-Level-1 (Blank Sample)		Cal-Level-2 (LLOQ)	
		0.0 ng/mL		1.0 ng/mL	
Run ID	**Sample No.**	**Peak Area [mV*min]**	**Acc. Criterion ≤ 20 % of LLOQ**	**Peak Area [mV*min]**	**Calculated 2439-exNA Conc. ng/mL**
AB01	3, 4	0.0018	7.6 %	0.0237	0.9
AB01	108, 109	0.0000	0.0 %	0.0254	1.0
AB02	3, 4	0.0000	0.0 %	0.0279	1.0
AB02	72, 73	0.0013	4.9 %	0.0263	0.9
AB03	3, 4	0.0000	0.0 %	0.0303	1.0
AB03	50, 51	0.0000	0.0 %	0.0298	1.0

Table 12: Selectivity for the Analysis of 2439-exNA in Rat CSF

					Blank Sample for Selectivity		QC-LLOQ		
					0.0 ng/mL		1.0 ng/mL		
Run ID	**Date of Analysis**	**Sex**	**Blank Matrix Lot No.**	**Sample No.**	**Peak Area [mV*min]**	**Acc. Criterion ≤ 20 % of LLOQ**	**Peak Area [mV*min]**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 20 % of Nominal**
AB01	18-Sep-2024	n.a.	RAT572673_28May2024	19, 22	0.0011	5.0 %	0.0219	0.9	90.0 %
AB01	18-Sep-2024	n.a.	RAT572674_28May2024	20, 23	0.0000	0.0 %	0.0229	0.9	90.0 %
AB01	18-Sep-2024	n.a.	RAT572675_28May2024	21, 24	0.0019	8.8 %	0.0216	0.8	80.0 %

Table 13: Matrix Effect for the Analysis of 2439-exNA in Rat CSF

			QC-Low		QC-High	
			3.0 ng/mL		900.0 ng/mL	
Run ID	**Blank Matrix Lot No.**	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**
AB01	RAT572673 _28May2024	26, 36	3.1	103.3 %	916.2	101.8 %
		27, 37	3.0	100.0 %	919.8	102.2 %
		28, 38	3.0	100.0 %	911.8	101.3 %
		Mean	3.03	101.1 %	915.93	101.8 %
		SD (±)	0.06		4.01	
		CV (%)	1.9 %		0.4 %	
		N	3		3	
AB01	RAT572674 _28May2024	29, 39	3.1	103.3 %	915.7	101.7 %
		30, 40	3.2	106.7 %	921.9	102.4 %
		31, 41	3.0	100.0 %	915.5	101.7 %
		Mean	3.10	103.3 %	917.70	102.0 %
		SD (±)	0.10		3.64	
		CV (%)	3.2 %		0.4 %	
		N	3		3	

Table 13: Matrix Effect for the Analysis of 2439-exNA in Rat CSF (Continued)

			QC-Low		QC-High	
			3.0 ng/mL		900.0 ng/mL	
Run ID	**Blank Matrix Lot No.**	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**
AB01	RAT572675 _28May2024	32, 42	3.0	100.0 %	915.7	101.7 %
		33, 43	3.3	110.0 %	915.7	101.7 %
		34, 44	3.1	103.3 %	923.1	102.6 %
		Mean	3.13	104.4 %	918.17	102.0 %
		SD (±)	0.15		4.27	
		CV (%)	4.9 %		0.5 %	
		N	3		3	

Table 14: Carry-Over for the Analysis of 2439-exNA in Artificial CSF

		Cal-Level-2 (LLOQ)			Cal-Level-7 (ULOQ)			Blank Sample for Carry-Over		
		1.0 ng/mL			1,000.0 ng/mL			0.0 ng/mL		
Run ID	**Sample No.**	**Peak Area [mV*min]**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 20 % of Nominal**	**Peak Area [mv*min]**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Peak Area [mV*min]**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ≤ 20 % of LLOQ**
AB01	4, 9, 10	0.0237	0.9	90.0 %	24.5453	989.4	98.9 %	0.0000	BQL	0.0 %
AB01	109, 114, 115	0.0254	1.0	100.0 %	24.9284	1,004.8	100.5 %	0.0000	BQL	0.0 %
AB02	4, 9, 10	0.0279	1.0	100.0 %	26.2594	983.7	98.4 %	0.0000	BQL	0.0 %
AB02	73, 78, 79	0.0263	0.9	90.0 %	26.8495	1,005.8	100.6 %	0.0024	BQL	9.1 %
AB03	4, 9, 10	0.0303	1.0	100.0 %	32.8381	987.6	98.8 %	0.0000	BQL	0.0 %
AB03	51, 56, 57	0.0298	1.0	100.0 %	33.6425	1,011.8	101.2 %	0.0000	BQL	0.0 %

Table 15: Results of the QC Samples (Accuracy and Precision) for the Analysis of 2439-exNA in Artificial CSF

					QC-LLOQ		QC-Low		QC-Mid		QC-High	
					1.0 ng/mL		3.0 ng/mL		450.0 ng/mL		900.0 ng/mL	
Run ID	**Date of Analysis**	**User**	**HPLC System**	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 20 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ±15 % of Nominal**
AB01	18-Sep-2024	LJ	HPLC-GLP-S-18	46, 53, 60, 67	0.8	80.0 %	3.1	103.3 %	466.5	103.7 %	922.1	102.5 %
				47, 54, 61, 68	0.8	80.0 %	3.0	100.0 %	468.7	104.2 %	924.8	102.8 %
				48, 55, 62, 69	0.8	80.0 %	3.0	100.0 %	465.0	103.3 %	928.3	103.1 %
				49, 56, 63, 70	0.9	90.0 %	3.0	100.0 %	465.9	103.5 %	921.5	102.4 %
				50, 57, 64, 71	0.8	80.0 %	3.1	103.3 %	470.5	104.6 %	923.1	102.6 %
				51, 58, 65, 72	0.9	90.0 %	3.1	103.3 %	466.2	103.6 %	924.4	102.7 %
Within-Run Accuracy/Precision I				**Mean**	0.83	83.3 %	3.05	101.7 %	467.13	103.8 %	924.03	102.7 %
				SD (±)	0.05		0.05		2.06		2.45	
				CV (%)	6.2 %		1.8 %		0.4 %		0.3 %	
				N	6		6		6		6	
AB02	20-Sep-2024	LJ	HPLC-GLP-S-18	19, 26, 33, 40	0.9	90.0 %	2.8	93.3 %	457.8	101.7 %	913.6	101.5 %
				20, 27, 34, 41	0.8	80.0 %	2.9	96.7 %	463.1	102.9 %	913.8	101.5 %
				21, 28, 35, 42	0.9	90.0 %	2.9	96.7 %	459.7	102.2 %	908.8	101.0 %
				22, 29, 36, 43	0.9	90.0 %	2.9	96.7 %	460.9	102.4 %	912.0	101.3 %
				23, 30, 37, 44	0.8	80.0 %	2.8	93.3 %	460.3	102.3 %	910.1	101.1 %
				24, 31, 38, 45	0.9	90.0 %	3.0	100.0 %	459.3	102.1 %	920.6	102.3 %
Within-Run Accuracy/Precision II				**Mean**	0.87	86.7 %	2.88	96.1 %	460.18	102.3 %	913.15	101.5 %
				SD (±)	0.05		0.08		1.77		4.14	
				CV (%)	6.0 %		2.6 %		0.4 %		0.5 %	
				N	6		6		6		6	

Table 15: Results of the QC Samples (Accuracy and Precision) for the Analysis of 2439-exNA in Artificial CSF (continued)

					QC-LLOQ		QC-Low		QC-Mid		QC-High	
					1.0 ng/mL		3.0 ng/mL		450.0 ng/mL		900.0 ng/mL	
Run ID	**Date of Analysis**	**User**	**HPLC System**	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 20 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ±15 % of Nominal**
AB03	25-Sep-2024	JSt	HPLC-GLP-7	12, 19, 26, 33	1.1	110.0 %	2.9	96.7 %	448.2	99.6 %	895.2	99.5 %
				13, 20, 27, 34	0.9	90.0 %	2.9	96.7 %	462.9	102.9 %	935.3	103.9 %
				14, 21, 28, 35	1.1	110.0 %	2.8	93.3 %	447.7	99.5 %	923.5	102.6 %
				15, 22, 29, 36	1.0	100.0 %	2.9	96.7 %	467.5	103.9 %	923.2	102.6 %
				16, 23, 30, 37	1.1	110.0 %	2.9	96.7 %	461.7	102.6 %	931.9	103.5 %
				17, 24, 31, 38	1.1	110.0 %	2.8	93.3 %	467.2	103.8 %	904.9	100.5 %
Within-Run Accuracy/Precision III				**Mean**	1.05	105.0 %	2.87	95.6 %	459.20	102.0 %	919.00	102.1 %
				SD (±)	0.08		0.05		9.01		15.72	
				CV (%)	8.0 %		1.8 %		2.0 %		1.7 %	
				N	6		6		6		6	
Between-Run Accuracy/Precision				**Mean**	0.92	91.7 %	2.93	97.8 %	462.17	102.7 %	918.73	102.1 %
				SD (±)	0.12		0.10		6.27		10.02	
				CV (%)	12.6 %		3.5 %		1.4 %		1.1 %	
				N	18		18		18		18	

Table 16: Dilution Integrity for the Analysis of 2439-exNA in Artificial CSF

		Dilution 1:1,000	
		500,000.0 ng/mL	
Run ID	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**
AB01	94	523,083.0	104.6 %
	95	517,465.2	103.5 %
	96	512,203.5	102.4 %
	97	521,800.7	104.4 %
	98	534,166.6	106.8 %
	99	517,654.8	103.5 %
	Mean	521,062.30	104.2 %
	SD (±)	7,475.02	
	CV (%)	1.4 %	
	N	6	

Table 17: Long-Term Stability for the Analysis of 2439-exNA in Artificial CSF

Run ID	Validation Parameters	Sample No.	QC-Low		QC-High		QC-500xULOQ (Dilution: 1:1,000)	
			3.0 ng/mL		900.0 ng/mL		500,000.0 ng/mL	
			Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal
AB01	T= 0 (-80 °C ± 20 °C) (Addition of Tween 20 / TCEP solution **after thawing;** storage **without** Tween 20 / TCEP)	74, 77, 80	2.8	93.3 %	894.2	99.4 %	500,323.9	100.1 %
		75, 78, 81	2.8	93.3 %	896.2	99.6 %	498,430.6	99.7 %
		76, 79, 82	2.8	93.3 %	891.6	99.1 %	492,417.6	98.5 %
		Mean	2.80	93.3 %	894.00	99.3 %	497,057.37	99.4 %
		SD (±)	0.00		2.31		4,128.16	
		CV (%)	0.0 %		0.3 %		0.8 %	
		N	3		3		3	
AB01	T= 0 (-80 °C ± 20 °C) (Addition of Tween 20 / TCEP solution **before freezing;** storage **with** Tween 20 / TCEP)	84, 87, 90	2.8	93.3 %	869.8	96.6 %	509,621.9	101.9 %
		85, 88, 91	2.9	96.7 %	873.6	97.1 %	503,285.6	100.7 %
		86, 89, 92	2.9	96.7 %	872.2	96.9 %	501,467.8	100.3 %
		Mean	2.87	95.6 %	871.87	96.9 %	504,791.77	101.0 %
		SD (±)	0.06		1.92		4,280.62	
		CV (%)	2.0 %		0.2 %		0.8 %	
		N	3		3		3	

Table 18: Freeze-Thaw and Short-Term Stability for the Analysis of 2439-exNA in Artificial CSF

Run ID	Validation Parameters	Sample No.	QC-Low		QC-High		QC-500xULOQ (Dilution: 1:1,000)	
			3.0 ng/mL		900.0 ng/mL		500,000.0 ng/mL	
			Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal
AB03	• FTS: 5 cycles (-80 °C ± 20 °C) • STS: 4 hours on ice	40, 43, 46	3.3	110.0 %	980.3	108.9 %	508,079.8	101.6 %
		41, 44, 47	3.2	106.7 %	980.5	108.9 %	483,731.4	96.7 %
		42, 45, 48	3.1	103.3 %	976.2	108.5 %	509,799.3	102.0 %
		Mean	3.20	106.7 %	979.00	108.8 %	500,536.83	100.1 %
		SD (±)	0.10		2.43		14,579.30	
		CV (%)	3.1 %		0.2 %		2.9 %	
		N	3		3		3	

Table 19: Bench-Top Stability / Robustness for the Analysis of 2439-exNA in Artificial CSF

Run ID	Stress Parameter	Sample No.	QC-500xULOQ (Dilution: 1:1,000) 500,000.0 ng/mL	
			Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal
AB02	3h at RT after dilution	59	439,142.3	87.8 %
		60	454,834.8	91.0 %
		61	453,912.6	90.8 %
		Mean	449,296.57	89.9 %
		SD (±)	8,805.93	
		CV (%)	2.0 %	
		N	3	

Table 20: Processed Sample Stability for the Analysis of 2439-exNA in Artificial CSF

Run ID	Date of Analysis	Sample No.	QC-Low 3.0 ng/mL		QC-High 900.0 ng/mL	
			Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal
AB01	18-Sep-2024 (Determination 1)	12, 15	3.0	100.0 %	912.7	101.4 %
		13, 16	3.0	100.0 %	913.6	101.5 %
		14, 17	3.0	100.0 %	916.9	101.9 %
		Mean	3.00	100.0 %	914.40	101.6 %
		SD (±)	0.00		2.21	
		CV (%)	0.0%		0.2 %	
		N	3		3	
AB01	19-Sep-2024 (Determination 2)	101, 104	3.1	103.3 %	923.7	102.6 %
		102, 105	3.1	103.3 %	921.5	102.4 %
		103, 106	3.0	100.0 %	918.0	102.0 %
		Mean	3.07	102.2 %	921.07	102.3 %
		SD (±)	0.06		2.87	
		CV (%)	1.9 %		0.3 %	
		N	3		3	

Table 20: Processed Sample Stability for the Analysis of 2439-exNA in Artificial CSF (Continued)

Run ID	Date of Analysis	Sample No.	QC-Low 3.0 ng/mL		QC-High 900.0 ng/mL	
			Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal
AB02	20-Sep-2024 (Determination 3)	12, 15	2.8	93.3 %	860.0	95.6 %
		13, 16	3.0	100.0 %	864.2	96.0 %
		14, 17	2.9	96.7 %	868.4	96.5 %
		Mean	2.90	96.7 %	864.20	96.0 %
		SD (±)	0.10		4.20	
		CV (%)	3.4 %		0.5 %	
		N	3		3	
AB02	21/22-Sep-2024 (Determination 4)	65, 68	3.0	100.0 %	870.2	96.7 %
		66, 69	2.9	96.7 %	855.8	95.1 %
		67, 70	2.8	93.3 %	867.9	96.4 %
		Mean	2.90	96.7 %	864.63	96.1 %
		SD (±)	0.10		7.74	
		CV (%)	3.4 %		0.9 %	
		N	3		3	

Table 21: Short-Term Stability of Stock and Working Solutions for the Analysis of 2439-exNA in Artificial CSF

Run ID	Validation Parameters	Sample No.	Working Solution LLOQ		Working Solution ULOQ		Stock Solution (diluted to QC-High)	
			1.0 ng/mL		1,000.0 ng/mL		900.0 ng/mL	
			Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 20 % of Nominal	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal
AB02	4 hours at room temperature	47, 51, 55	0.9	90.0 %	998.2	99.8 %	847.4	94.2%
		48, 52, 56	0.9	90.0 %	993.4	99.3 %	845.7	94.0 %
		49, 53, 57	0.9	90.0 %	977.0	97.7 %	845.8	94.0 %
		Mean	0.90	90.0 %	989.53	99.0 %	846.30	94.0 %
		SD (±)	0.00		11.12		0.95	
		CV (%)	0.0 %		1.1 %		0.1 %	
		N	3		3		3	

14 SUPPLEMENTARY INFORMATION

14.1 Supplemental File 1

Axolabs *MDR_2439exNA_CSF*

14.2 Supplemental File 2

Axolabs *STM_Broad_2439-exNA_RatCSF*